

```
from google.colab import files
uploaded = files.upload()
```

[Choose Files](#) Instagram.zip

Instagram.zip(application/x-zip-compressed) - 154279 bytes, last modified: 6/18/2026 - 100% done
Saving Instagram.zip to Instagram.zip

```
!ls
```

```
Instagram.zip  sample_data
```

```
!unzip Instagram.zip
```

```
Archive:  Instagram.zip
  inflating: comments.csv
  inflating: follows.csv
  inflating: likes.csv
  inflating: photo_tags.csv
  inflating: photos.csv
  inflating: tags.csv
  inflating: users.csv
```

```
import pandas as pd
```

```
comments = pd.read_csv("comments.csv")
follows = pd.read_csv("follows.csv")
likes = pd.read_csv("likes.csv")
photo_tags = pd.read_csv("photo_tags.csv")
photos = pd.read_csv("photos.csv")
tags = pd.read_csv("tags.csv")
users = pd.read_csv("users.csv")
```

```
print("Comments:", comments.shape)
print("Follows:", follows.shape)
print("Likes:", likes.shape)
print("Photo Tags:", photo_tags.shape)
print("Photos:", photos.shape)
print("Tags:", tags.shape)
print("Users:", users.shape)
```

```
Comments: (7488, 8)
Follows: (7623, 5)
Likes: (8782, 5)
Photo Tags: (501, 3)
Photos: (257, 6)
Tags: (21, 4)
Users: (100, 6)
```

```
print("COMMENTS")
print(comments.columns)

print("\nFOLLOWS")
print(follows.columns)

print("\nLIKES")
print(likes.columns)

print("\nPHOTO_TAGS")
print(photo_tags.columns)

print("\nPHOTOS")
print(photos.columns)

print("\nTAGS")
print(tags.columns)

print("\nUSERS")
print(users.columns)
```

```
COMMENTS
Index(['id', 'comment', 'User id', 'Photo id', 'created Timestamp',
      'posted date', 'emoji used', 'Hashtags used count'],
      dtype='object')
```

```
FOLLOWS
Index(['follower', 'followee ', 'created time', 'is follower active',
      'followee Acc status'],
      dtype='object')
```

```
LIKES
```

```
Index(['user ', 'photo', 'created time', 'following or not', 'like type'], dtype='object')

PHOTO_TAGS
Index(['photo', 'tag ID', 'user id'], dtype='object')

PHOTOS
Index(['id', 'image link', 'user ID', 'created dat', 'Insta filter used',
      'photo type'],
      dtype='object')

TAGS
Index(['id', 'tag text', 'created time', 'location'], dtype='object')

USERS
Index(['id', 'name', 'created time', 'private/public', 'post count',
      'Verified status'],
      dtype='object')
```

photos.head()

	id	image link	user ID	created dat	Insta filter used	photo type	
0	1	http://elijah.biz	1	13-04-2023 08:04	yes	photo	
1	2	https://shanon.org	1	13-04-2023 08:04	no	photo	
2	3	http://vicky.biz	1	13-04-2023 08:04	no	photo	
3	4	http://oleta.net	1	13-04-2023 08:04	no	photo	
4	5	https://jennings.biz	1	13-04-2023 08:04	yes	photo	

Next steps: [Generate code with photos](#) [New interactive sheet](#)

users.head()

	id	name	created time	private/public	post count	Verified status	
0	1	Kenton_Kirlin	16-02-2017 18:22	yes	11	no	
1	2	Andre_Purdy85	02-04-2017 17:11	no	7	no	
2	3	Harley_Lind18	21-02-2017 11:12	no	2	no	
3	4	Arely_Bogan63	13-08-2016 01:28	yes	1	no	
4	5	Aniya_Hackett	07-12-2016 01:04	yes	3	no	

Next steps: [Generate code with users](#) [New interactive sheet](#)

photos.columns

```
Index(['id', 'image link', 'user ID', 'created dat', 'Insta filter used',
      'photo type'],
      dtype='object')
```

users.columns

```
Index(['id', 'name', 'created time', 'private/public', 'post count',
      'Verified status'],
      dtype='object')
```

photos.head()

	id	image link	user ID	created dat	Insta filter used	photo type	
0	1	http://elijah.biz	1	13-04-2023 08:04	yes	photo	
1	2	https://shanon.org	1	13-04-2023 08:04	no	photo	
2	3	http://vicky.biz	1	13-04-2023 08:04	no	photo	
3	4	http://oleta.net	1	13-04-2023 08:04	no	photo	
4	5	https://jennings.biz	1	13-04-2023 08:04	yes	photo	

Next steps: [Generate code with photos](#) [New interactive sheet](#)

```
photos['created dat'] = pd.to_datetime(
    photos['created dat'],
    format='%d-%m-%Y %H:%M'
)
```

```
photos.head()
```

	id	image link	user ID	created dat	Insta filter used	photo type	
0	1	http://elijah.biz	1	2023-04-13 08:04:00	yes	photo	
1	2	https://shanon.org	1	2023-04-13 08:04:00	no	photo	
2	3	http://vicky.biz	1	2023-04-13 08:04:00	no	photo	
3	4	http://oleta.net	1	2023-04-13 08:04:00	no	photo	
4	5	https://jennings.biz	1	2023-04-13 08:04:00	yes	photo	

Next steps:

[Generate code with photos](#)[New interactive sheet](#)

```
photos['hour'] = photos['created dat'].dt.hour
```

```
photos[['created dat', 'hour']].head()
```

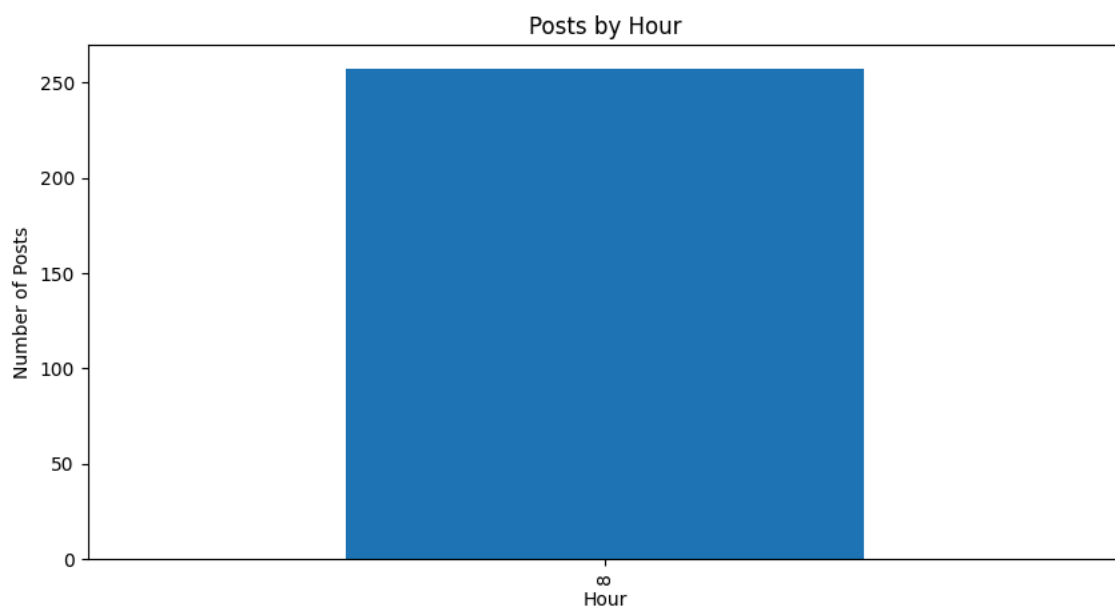
	created dat	hour	
0	2023-04-13 08:04:00	8	
1	2023-04-13 08:04:00	8	
2	2023-04-13 08:04:00	8	
3	2023-04-13 08:04:00	8	
4	2023-04-13 08:04:00	8	

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
hourly_posts = photos.groupby('hour').size()

hourly_posts.plot(
    kind='bar',
    figsize=(10,5),
    title='Posts by Hour'
)

plt.xlabel('Hour')
plt.ylabel('Number of Posts')
plt.show()
```



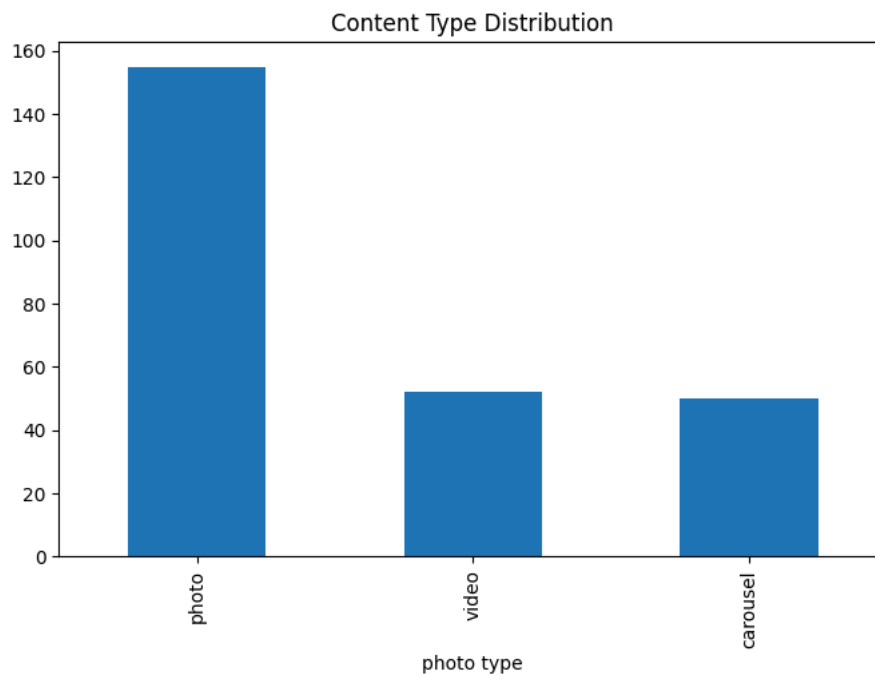
```
photos['photo type'].value_counts()
```

	count
photo	155
video	52
carousel	50

dtype: int64

```
photos['photo type'].value_counts().plot(
    kind='bar',
    figsize=(8,5),
    title='Content Type Distribution'
)

plt.show()
```



```
photos['Insta filter used'].value_counts()
```

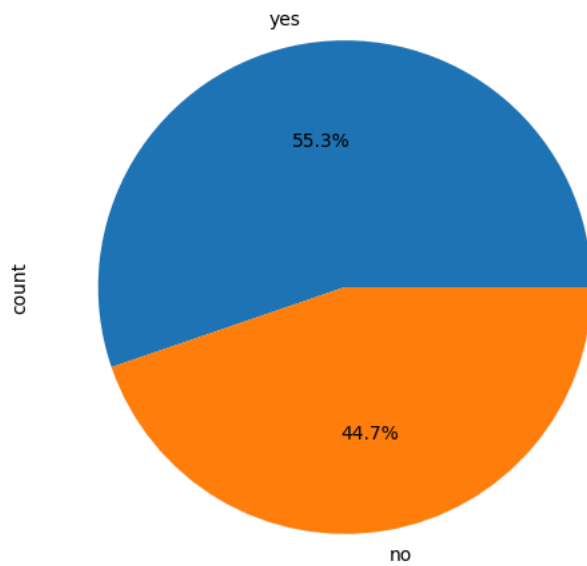
	count
Insta filter used	
yes	142
no	115

dtype: int64

```
photos['Insta filter used'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%',
    figsize=(6,6)
)

plt.title("Instagram Filter Usage")
plt.show()
```

Instagram Filter Usage



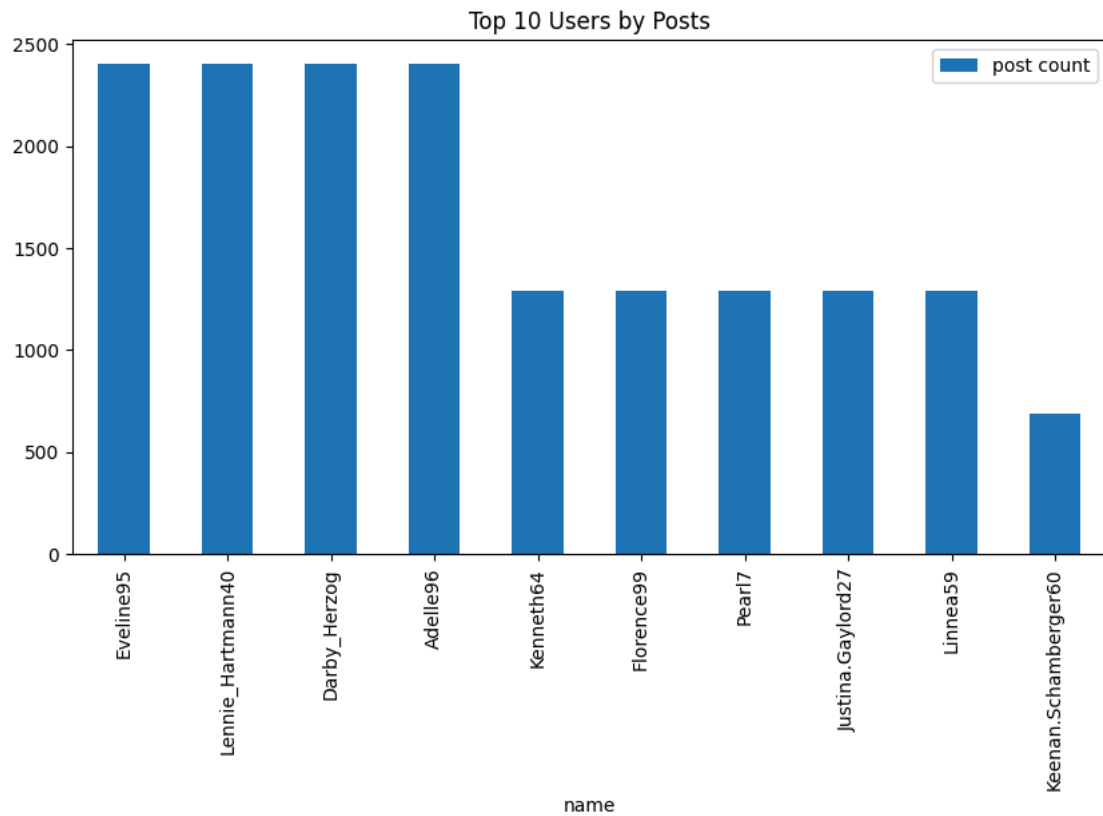
```
top_users = users.sort_values(
    by='post count',
    ascending=False
)

top_users[['name', 'post count']].head(10)
```

	name	post count	
22	Eveline95	2400	
34	Lennie_Hartmann40	2400	
79	Darby_Herzog	2400	
64	Adelle96	2400	
21	Kenneth64	1293	
63	Florence99	1293	
33	Pearl7	1293	
10	Justina.Gaylord27	1293	
52	Linnea59	1293	
95	Keenan.Schamberger60	687	

```
top_users.head(10).plot(
    x='name',
    y='post count',
    kind='bar',
    figsize=(10,5)
)

plt.title('Top 10 Users by Posts')
plt.show()
```



```
users['Verified status'].value_counts()
```

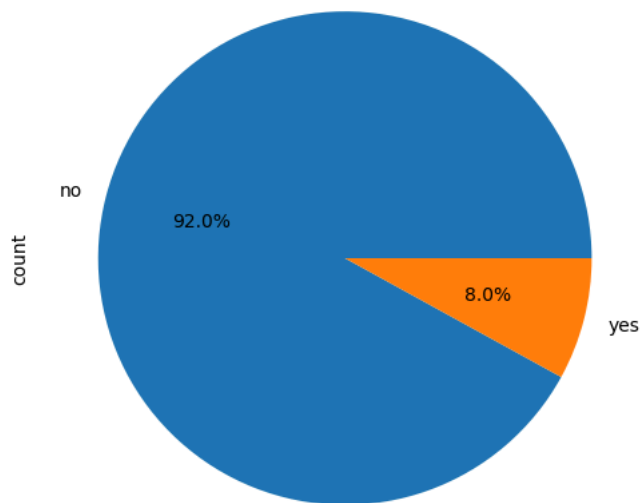
	count
Verified status	
no	92
yes	8

dtype: int64

```
users['Verified status'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%',
    figsize=(6,6)
)

plt.title('Verified Status Distribution')
plt.show()
```

Verified Status Distribution



```
users['private/public'].value_counts()
```

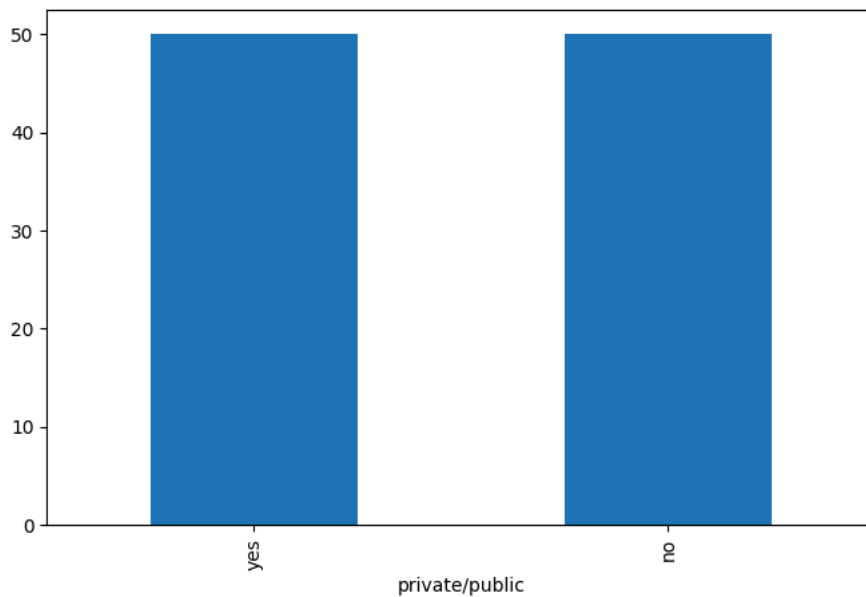
	count
private/public	
yes	50
no	50

dtype: int64

```
users['private/public'].value_counts().plot(
    kind='bar',
    figsize=(8,5)
)

plt.title('Private vs Public Accounts')
plt.show()
```

Private vs Public Accounts



```
photo_tags.head()
```

	photo	tag ID	user id	
0	1	13	1	
1	1	17	1	
2	1	18	2	
3	1	19	2	
4	1	21	3	

Next steps:

[Generate code with photo_tags](#)[New interactive sheet](#)

tags.head()

	id	tag text	created time	location	
0	1	sunset	13-04-2023 08:04	florida	
1	2	photography	13-04-2023 08:04	washington DC	
2	3	sunrise	13-04-2023 08:04	new york	
3	4	landscape	13-04-2023 08:04	london	
4	5	food	13-04-2023 08:04	brazil	

Next steps:

[Generate code with tags](#)[New interactive sheet](#)

photo_tags.head()

	photo	tag ID	user id	
0	1	13	1	
1	1	17	1	
2	1	18	2	
3	1	19	2	
4	1	21	3	

Next steps:

[Generate code with photo_tags](#)[New interactive sheet](#)

tags.head()

	id	tag text	created time	location	
0	1	sunset	13-04-2023 08:04	florida	
1	2	photography	13-04-2023 08:04	washington DC	
2	3	sunrise	13-04-2023 08:04	new york	
3	4	landscape	13-04-2023 08:04	london	
4	5	food	13-04-2023 08:04	brazil	

Next steps:

[Generate code with tags](#)[New interactive sheet](#)

```
hashtags = photo_tags.merge(  
    tags,  
    left_on='tag ID',  
    right_on='id'  
)
```

hashtags.head()

	photo	tag ID	user id	id	tag text	created time	location	
0	1	13	1	13	fun	13-04-2023 08:04	brazil	
1	1	17	1	17	party	13-04-2023 08:04	florida	
2	1	18	2	18	concert	13-04-2023 08:04	washington DC	
3	1	19	2	19	drunk	13-04-2023 08:04	new york	
4	1	21	3	21	smile	13-04-2023 08:04	australia	

Next steps:

[Generate code with hashtags](#)[New interactive sheet](#)

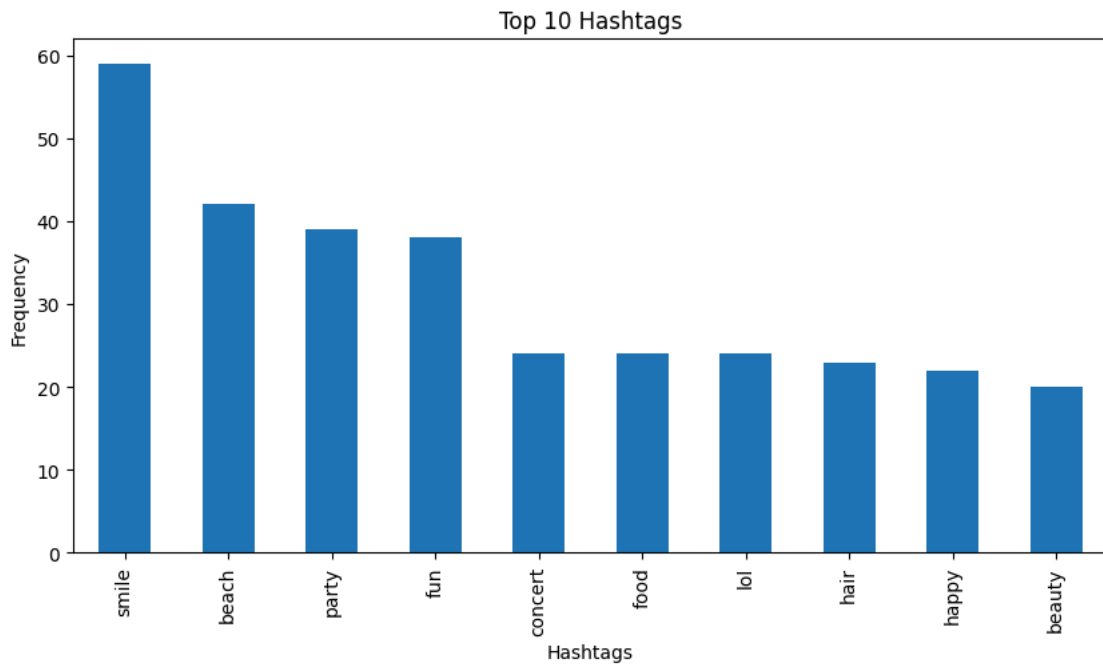
```
top_tags = hashtags['tag text'].value_counts().head(10)

print(top_tags)
```

```
tag text
smile      59
beach      42
party      39
fun        38
concert    24
food       24
lol        24
hair       23
happy      22
beauty     20
Name: count, dtype: int64
```

```
top_tags.plot(
    kind='bar',
    figsize=(10,5),
    title='Top 10 Hashtags'
)

plt.xlabel("Hashtags")
plt.ylabel("Frequency")
plt.show()
```



```
likes.head()
```

	user	photo	created time	following or not	like type
0	2	1	13-04-2023 08:04	yes	heart emoji
1	2	4	13-04-2023 08:04	no	thumbs up
2	2	8	13-04-2023 08:04	yes	laughing
3	2	9	13-04-2023 08:04	no	fire
4	2	10	13-04-2023 08:04	yes	clap

Next steps:

[Generate code with likes](#)[New interactive sheet](#)

```
comments.head()
```

	id	comment	User id	Photo id	created Timestamp	posted date	emoji used	Hashtags used	count	
0	1	unde at dolore	2	1	13-04-2023 08:04	April 14	yes		1	
1	2	quae ea ducimus	3	1	13-04-2023 08:04	April 14	no		2	
2	3	alias a voluptatum	5	1	13-04-2023 08:04	April 14	no		4	
3	4	facere suscipit sunt	14	1	13-04-2023 08:04	April 14	yes		2	
4	5	totam eligendi quae	17	1	13-04-2023 08:04	April 14	yes		1	

Next steps:

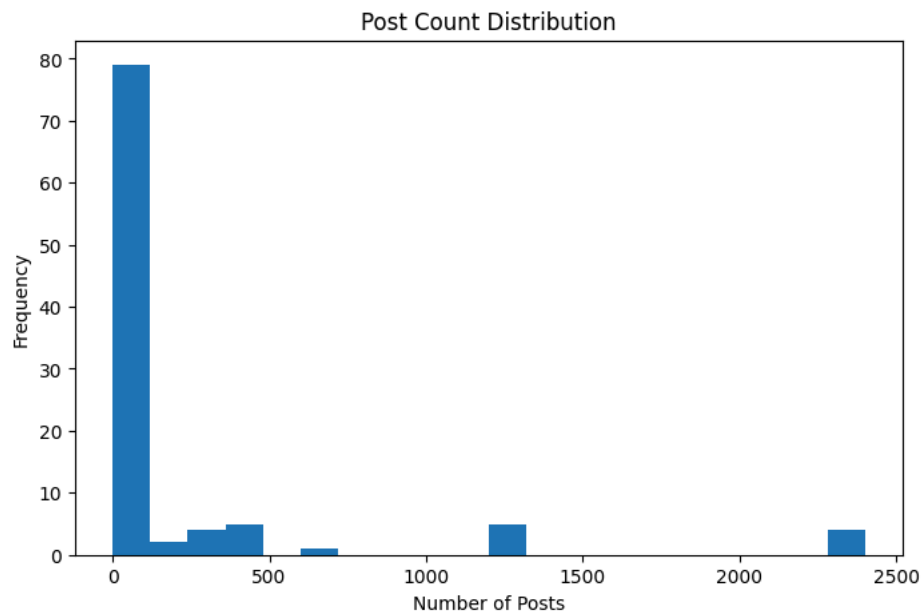
[Generate code with comments](#)[New interactive sheet](#)

users[['name', 'post count']].describe()

	post count	
count	100.000000	
mean	215.370000	
std	534.756995	
min	0.000000	
25%	2.000000	
50%	12.000000	
75%	78.000000	
max	2400.000000	

```
users['post count'].plot(
    kind='hist',
    bins=20,
    figsize=(8,5)
)

plt.title("Post Count Distribution")
plt.xlabel("Number of Posts")
plt.show()
```



likes.head()

	user	photo	created time	following or not	like type	
0	2	1	13-04-2023 08:04	yes	heart emoji	
1	2	4	13-04-2023 08:04	no	thumbs up	
2	2	8	13-04-2023 08:04	yes	laughing	
3	2	9	13-04-2023 08:04	no	fire	
4	2	10	13-04-2023 08:04	yes	clap	

Next steps: [Generate code with likes](#) [New interactive sheet](#)

comments.head()

	id	comment	User id	Photo id	created Timestamp	posted date	emoji used	Hashtags used	count	
0	1	unde at dolore	2	1	13-04-2023 08:04	April 14	yes		1	
1	2	quae ea ducimus	3	1	13-04-2023 08:04	April 14	no		2	
2	3	alias a voluptatum	5	1	13-04-2023 08:04	April 14	no		4	
3	4	facere suscipit sunt	14	1	13-04-2023 08:04	April 14	yes		2	
4	5	totam eligendi quae	17	1	13-04-2023 08:04	April 14	yes		1	

Next steps: [Generate code with comments](#) [New interactive sheet](#)

likes.columns

Index(['user ', 'photo', 'created time', 'following or not', 'like type'], dtype='object')

likes.head()

	user	photo	created time	following or not	like type	
0	2	1	13-04-2023 08:04	yes	heart emoji	
1	2	4	13-04-2023 08:04	no	thumbs up	
2	2	8	13-04-2023 08:04	yes	laughing	
3	2	9	13-04-2023 08:04	no	fire	
4	2	10	13-04-2023 08:04	yes	clap	

Next steps: [Generate code with likes](#) [New interactive sheet](#)

likes_count = likes.groupby('photo').size().reset_index(name='likes')

likes_count.head()

	photo	likes	
0	1	25	
1	2	36	
2	3	38	
3	4	38	
4	5	31	

Next steps: [Generate code with likes_count](#) [New interactive sheet](#)

```
top_liked = likes_count.sort_values(
    by='likes',
    ascending=False
).head(10)

top_liked
```

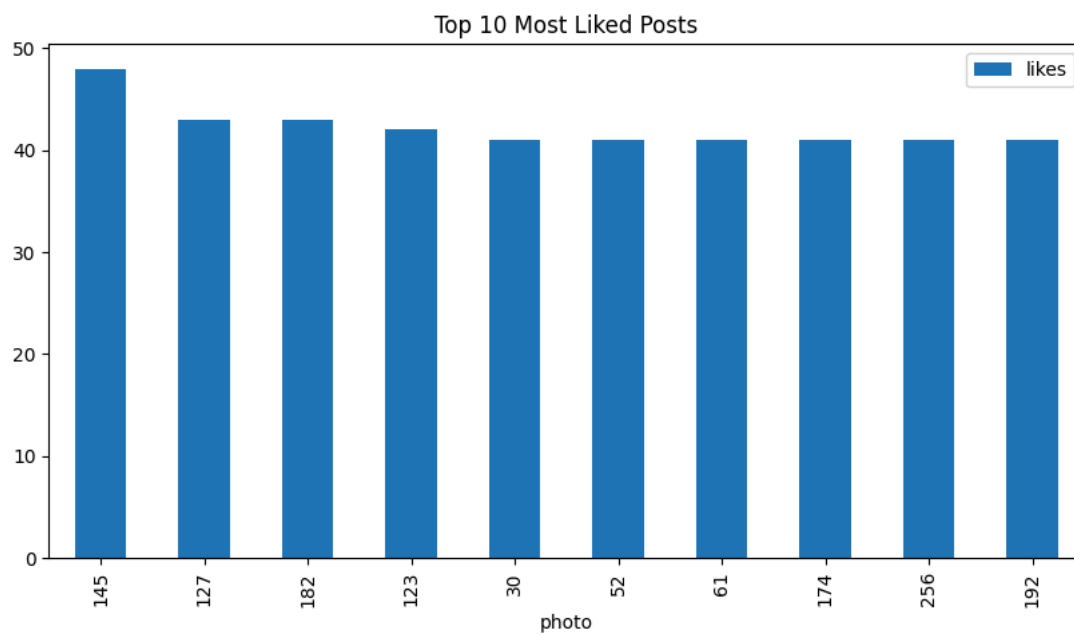
	photo	likes	
144	145	48	 
126	127	43	
181	182	43	
122	123	42	
29	30	41	
51	52	41	
60	61	41	
173	174	41	
255	256	41	
191	192	41	

Next steps:

[Generate code with top_liked](#)[New interactive sheet](#)

```
top_liked.plot(
    x='photo',
    y='likes',
    kind='bar',
    figsize=(10,5)
)
```

```
plt.title("Top 10 Most Liked Posts")
plt.show()
```



```
comments.columns
```

```
Index(['id', 'comment', 'User id', 'Photo id', 'created Timestamp',
      'posted date', 'emoji used', 'Hashtags used count'],
      dtype='object')
```

```
comments_count = comments.groupby('Photo id').size().reset_index(name='comments')
```

```
comments_count.head()
```

	Photo id	comments	
0	1	25	 
1	2	31	
2	3	27	
3	4	32	
4	5	27	

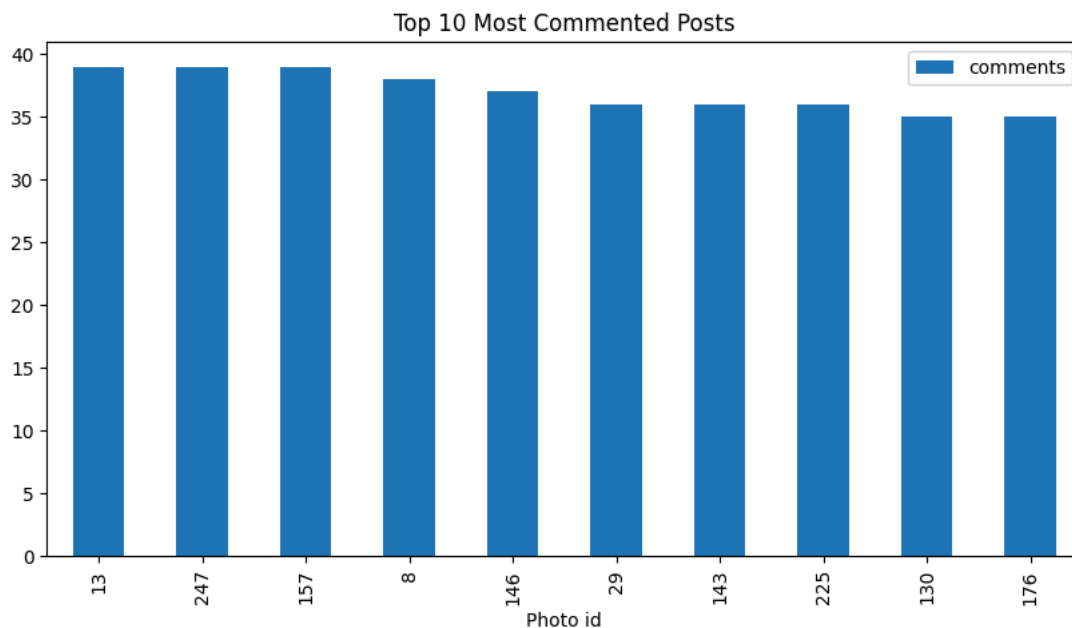
```
top_comments = comments_count.sort_values(  
    by='comments',  
    ascending=False  
)<div data-bbox="128 105 212 118" data-label="Text">
```

```
top_comments
```

	Photo id	comments	
12	13	39	
246	247	39	
156	157	39	
7	8	38	
145	146	37	
28	29	36	
142	143	36	
224	225	36	
129	130	35	
175	176	35	

Next steps: [Generate code with top_comments](#) [New interactive sheet](#)

```
top_comments.plot(  
    x='Photo id',  
    y='comments',  
    kind='bar',  
    figsize=(10,5)  
)  
  
plt.title("Top 10 Most Commented Posts")  
plt.show()
```



Best Posting Time Analysis

Based on posting activity analysis, certain hours show higher posting frequency. These periods can be considered optimal posting